

# HACKATHON 2.0 - Week 2 Progress Report

## Knowvia.

Team: The Algorithm | 17<sup>th</sup> September, 2026

**GitHub:** [github.com/Sydonverse/Knowvia](https://github.com/Sydonverse/Knowvia)

**Demo video:** [Click Here](#)

### Week 2 Focus

Week 1 we focused on planning the PRD, data model, and architecture. This week we moved into implementation: the department-scoped backend, the assignment lifecycle (creation, submission, review), class scheduling with automated reminders, and file-upload safety checks. This report shows one complete working process and reports status against every objective in the PRD.

### A. Objectives Status & Evidence

| Objective                                                                      | Status      | Week 2 Progress / Evidence                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 — Class schedule shown as the primary dashboard view                        | In Progress | Scheduling engine and an hourly automated reminder job are implemented on the backend. Dashboard-first display is still being wired into the client UI.                                                                                    |
| 02 — Materials uploaded with type/size validation and malware screening        | In Progress | Upload validation is implemented: file signatures are checked against declared type, and disallowed extensions are blocked before processing. Files are currently held in server memory pending permanent cloud storage — not yet durable. |
| 03 — Assignment creation auto-generates an announcement linking to full detail | Achieved    | Shown in the demo video: creating an assignment generates an announcement that opens directly to the assignment record.                                                                                                                    |
| 04 — Student submission updates the dashboard progress meter                   | Achieved    | Shown in the demo video: submitting and having work approved moves the intern's progress meter and status.                                                                                                                                 |

## B. Significant Changes From the PRD

| What Changed                                                                                                                                                                       | Why It Was Changed                                                                                                                                   | Impact on Project / Objectives                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Backend moved from Supabase-native services (Auth, Storage, Realtime) to a custom Node.js/Express API with Prisma ORM and Socket.io, using Supabase only as the Postgres database. | The custom API gives finer control over role-based access rules and the multi-state assignment review flow than the managed services alone provided. | No objective changes. Architecture and technology choice only — department segregation and every stated objective remain the same. |

## C. Current Challenges & Proposed Solutions

| Current Challenge                                                             | Effect on Development                                                                                            | Proposed Solution / Next Action                                                            |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Uploaded files are held in server memory rather than permanent cloud storage. | Files do not yet persist reliably across server restarts — materials and submissions are not production-durable. | Connect live Supabase Storage credentials this week so uploads write to permanent storage. |
| Offline access is limited to the app shell.                                   | Interns without a stable connection cannot yet view previously loaded learning materials.                        | Extend service worker caching to support read-only offline access to materials.            |